

Nika Ghaderi

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Education

- **B.Sc. in Computer Engineering, Sharif University of Technology, Tehran, Iran** *2022 – Expected 2027*
GPA: 19.30/20
- **Farzanegan High School (SAMPAD), Iran** *2022*
Mathematics and Physics track, total average: 19.96/20

Research Experience

- **Undergraduate Researcher (B.Sc. Thesis), DAISY Lab - Supervisor: Dr. Maryam Ramezani** *March 2026 – Present*
Sharif University of Technology, Tehran, Iran
Learning Query Cost Models that Extrapolate Across Data Scale
 - Extending a graph-transformer query-cost model to predict Spark SQL latency and I/O from compile-time query plans, with operator weights conditioned on each operator's type and its parent's type.
 - Developing methods to predict costs at 100× the training data size without retraining: scale-free plan features, a power-law readout with a learned growth exponent, and a symbolic analysis of how each operator's cardinality grows with data size.
 - Built a two-server pipeline collecting isolated Spark execution traces on TPC-H at multiple scale factors.
 - Preliminary: the relation-aware encoder lowers I/O error by ~57% vs. a shared-weight baseline; scale-free features cut extrapolation error ~4× in a 10× scale-up pilot.
- **Software Engineering Intern, OpenAirInterface 5G Project** *July – September 2025*
EURECOM, Sophia Antipolis, France - [GitLab](#)
 - Spearheaded GPU acceleration for 5G NR channel simulations, offloading multipath convolution and noise generation to CUDA pipelines. Over 400x speedup vs. a single-threaded scalar C baseline.
 - Optimized CUDA kernels via tiled convolution with shared memory to maximize data reuse and cut global memory traffic; used cuRAND for massively parallel noise generation.
 - Profiled and benchmarked three CUDA memory models (Explicit Copy, Unified Memory, ATS), selecting the optimal model per use case from performance data.
 - Built a CI-integrated CTest benchmark suite for numerical correctness and regression testing.
- **AI Research Assistant, Online Learning Project** *January – March 2024*
Dr. Rohban's AI & Robotics Lab, Sharif University of Technology
 - Researched online learning methods for adaptive AI systems in dynamic environments under the direct supervision of Dr. Sajad Sabzi, developing and evaluating real-time learning algorithms.

Professional Experience

- **C/C++ Software Engineer (Full-time)** *February 2026 – Present*
Sharif HPC Center, Sharif University of Technology
Real-time deep packet inspection and call-detail-record correlation for high-throughput VoIP networks.
 - Designed a GPU-accelerated multi-pattern matching engine in CUDA for VoIP signaling detection (SIP/RTP), replacing a production Intel Hyperscan CPU implementation → 100 Gbps end-to-end vs. a 10–20 Gbps 8-thread CPU baseline.
 - Developed a heuristic backward-scan algorithm identifying encapsulated protocol layers in tunneled traffic via header-signature and codec-field validation, cutting the false-positive rate among candidate matches from ~25% to none observed on validation traffic; bounded scan depth to reduce GPU warp divergence.

- Replaced a lock-based allocator with a lock-free, sequence-number-based ring buffer supporting out-of-order packet placement under burst traffic, eliminating hot-path contention; debugged race conditions and stale-buffer defects with cuda-gdb and Python golden-model validation.
- Built a lock-free multi-threaded producer–consumer pipeline in C++17 (atomic CAS object pool, MPMC queue) for CDR correlation with zero per-message heap allocation, plus a multi-threaded Kafka layer (librdkafka) and a deterministic offline replay transport preserving original inter-message timing.
- Researched shared-memory hash table architectures (DPDK cuckoo-hashing `rte_hash`, seqlock-based flat arrays) for a correlation state table sized for 10M+ concurrent flows.

Teaching and Academic Service

- **Teaching Assistant, Sharif University of Technology** 2023 – 2026
Designed and contributed to homework, quizzes, and exams for: Linear Algebra (*Fa26, Sp26, Sp25, Fa25, Sp23, Fa23*) - Artificial Intelligence (*Fa26, Sp26, Fa25*) - Computer Networks (*Fa26*) - Compiler Design (*Fa26, Sp26*) - Scientific and Technical Presentation (*Sp26, Fa25*) - Logic Circuits (*Sp26, Sp25, Fa24*) - Database Design (*Sp26, Sp25*) - Formal Languages and Automata (*Sp26, Sp25*) - Game Theory (*Sp26*) - Computer Architecture (*Fa25*) - Fundamentals of Programming in C (*Fa24*)
- **Grader, Iranian National Olympiad in Artificial Intelligence, Young Scholars Club (YSC)** 2026
Graded papers determining the national medalists in the selection for the International Olympiad in AI (IOAI).

Projects

- [Linux Container Runtime Development](#) 2025
Built a lightweight, daemonless container runtime from scratch in C: filesystem isolation via chroot, process and mount isolation via Linux namespaces (PID, mount), CPU and memory limits enforced with cgroups v2, and a custom CLI with an automated setup script.
- [64B/66B to 8B/10B Encoding Converter with Error Correction](#) 2025
Built a real-time Arduino-based converter from 64B/66B to 8B/10B encoding, with Hamming-code error detection and correction, plus simulated fault injection for resilience testing. — [Demo](#)
- [Game Development Project](#) 2023
Designed a strategy game inspired by Stronghold Crusader using the Java programming language. — [Demo](#)
- [File Editing Tool Development](#) 2022
Developed a Vim-inspired text editor in C for an “Introduction to Programming” course.

Academic Programs and Training

- **Artificial Intelligence and Digital Technologies, SPbPU (St. Petersburg Polytechnic)** 2025
Fully funded; selected to represent Sharif University of Technology under a bilateral inter-university agreement. Completed the “Innovator” track at the Winter University of Engineering Sciences, SPbPU, Russia.
- **Neural Networks and Deep Learning, Coursera (DeepLearning.AI)** — [Credential](#) 2025
- **Front-End Training Course, Quera College — Perfect Score** 2023

Honors and Awards

- 2022: Ranked 23rd of 52,009 (Region 2) in the Mathematics track, Iranian Nationwide University Entrance Exam (Concours); top 0.05% nationally among 145,657 candidates.
- 2020: Awarded a gold medal in the 11th-grade *Tehran Math House competition*, Iran.
- 2020: Placed among Iran's top 5 in the Cayley Contest, organized by *CEMC, University of Waterloo, Canada*.
- 2017: Awarded the Top Scholar plaque by the *Association of Young Mathematicians*, Iran.

Foreign Languages

- Persian: Native | English: C1, Advanced | French: B1, Intermediate (CEFR)